

Then everything crashed. Isutani had bought Pebble Beach a year after the Nikkei had hit its peak in 1989, and he was later investigated for money laundering by the FBI.⁵ Figure 3.1 shows that since 1990 Japanese stocks have been flat. But while Japan was languishing, the United States boomed. Even so, Japanese cumulated returns were higher at December 2011 than U.S. cumulated returns. Since 2000, Figure 3.1 shows that the United States and Japan have a greater tendency to move together. They jointly slowed during the early 2000s, experienced bull markets during the mid-2000s, and then crashed during the financial crisis of 2007–2008. Over the whole sample, however, Japan has moved very differently from the United States.

The average return and volatility for the United States in Figure 3.1 are 10.3% and 15.7%, respectively. The corresponding numbers for Japan are 11.1% and 21.7%, respectively. We plot these rewards (means) and risks (volatilities) in mean-standard deviation space in Figure 3.2. The United States is represented by the square, and Japan is represented by the circle. The x -axis is in standard deviation units, and the y -axis units are average returns.

The curve linking the United States and Japan in Figure 3.2 is the *mean-variance frontier*. Like the literature, I use the terms mean-variance frontier and mean-standard deviation frontier interchangeably as the two can be obtained simply by squaring, or taking a square root of, the x -axis depending on whether one

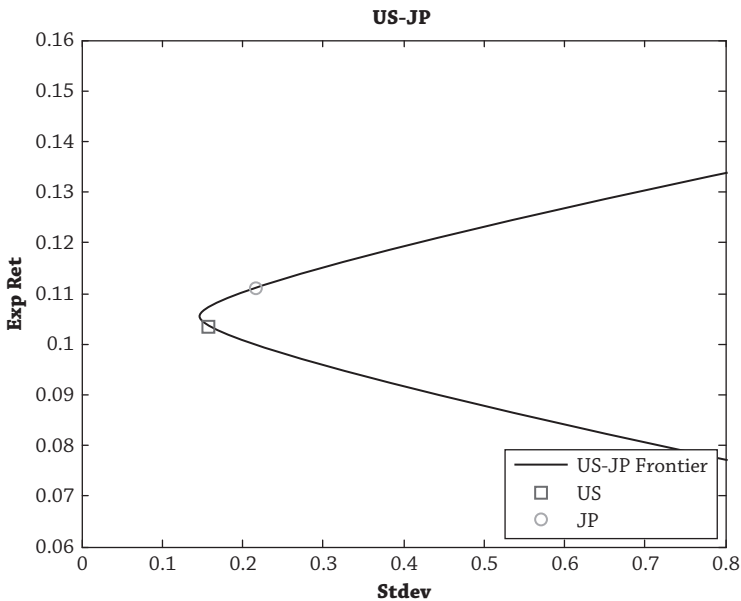


Figure 3.2

⁵“A Japanese Laundry Worth \$1 Billion?” *Businessweek*, May 24, 1993.